

THE CHINESE UNIVERSITY OF HONG KONG

Design a Terahertz Photonic Crystal Cavity

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Abstract

Photonic crystals(PC) are periodically arranged dielectric materials, by controlling the structure, size , dielectric constants of it, we can confine EM-wave as specific position as modulation, as a waveguide that the light are well guide and couple to certain positions. Leading photonic crystals has a high potential on its application such as optic signal modulation, spectroscopic sensing , ultra-broadband wireless communication and laser technology. In this project, we will focus on design a 1 dimensional photonic crystal cavity in the range of Terahertz frequency. Explain the theory of EM wave propagate in the photonic crystal, the method use in the simulation. We will use Lumerical FDTD software to demonstrate the design of photonic crystal structure and the setting of parameters, analysis the performance and the goal is to achieve a high quality factor of the cavity.

Acknowledgements

I am very grateful to my supervisor Professor Sun XianKai, and provide me a chance on the research of the photonic field, guidance on my final year project, the group meeting also let me learn about much more photonic technology other than the fyp topic and the valuable discussions. Also I would like to thank Professor Tsang HonKi as my Associate Examiner to give comments and suggestion on my project and presentations. At last I would like to thank TA group, Xi Xiang and Ma Jingwen's fruitful discussions and suggestions.

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Chapter 1.

Introduction

Photonic crystal are develop to control light propagation, leading a wide range of application. The components which are arrange in a periodic pattern are call crystal in which can be extend from one dimension to three dimensions. By the differing the material dielectric constant, patterns, side and distance, we can guide (waveguide) and trap (cavity) the EM-wave with a small loss. For a two dimensional planer photonic crystal cavity. One kind of 2D PCs structure is dielectric slab with 2D air hole are widely used due to the strong light confinement and simple fabrication. The EM- wave will guide in x-y direction and trap in z direction, by plotting the band structure of the crystal in which can visualize the spectrum of mode which is possible to sustain for a long time and decay extremely slow, for 2D PC with band gap, within the bandgap, no mode are allowed, by setting a single lattice site, we can create a single localized mode or a set of closely spaced modes frequencies within the gap. Beside Terahertz wave are one of most interest research in the past decades, it locate in the range between radio and light waves 0.1 to 10THz,with its high potential applications such as sensing, communication, demultiplexing, and produce high performance device. High Q-factor are also the interest of research.

Photonic-Crystal with Terahertz Frequency

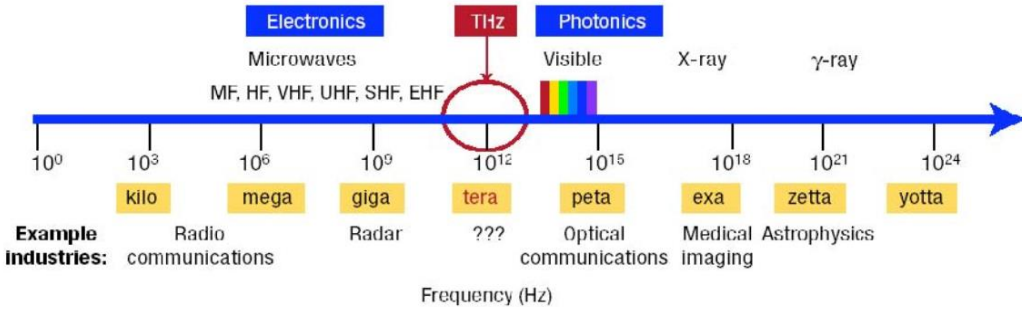


Fig1.1

The terahertz range is define 0.1 to 10THz,In my project we will focus on 0.3 to 3THz

Photonic-Crystal

The dielectric constant also periodic ε(r)= ε(r+ R),which EM wave propagration govern by Bloch's theorem. $\mathbf{H}_{(n,k_z,\mathbf{k}_\parallel)}(\mathbf{r}) = e^{i\mathbf{k}_\parallel\cdot\rho}e^{ik_z z}\mathbf{u}_{(n,k_z,\mathbf{k}_\parallel)}(\boldsymbol{\rho})$, the tern u(p) is the structural pattern related to dielectric constant.

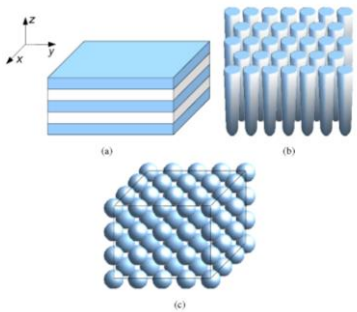


Fig1.2

A crystal is a material whose components (atoms, molecules, or ions) are arranged in an orderly repeating pattern extending in one, two, or three dimensions.Here is the graph demonstrate a triangular lattice and square array structure 2.

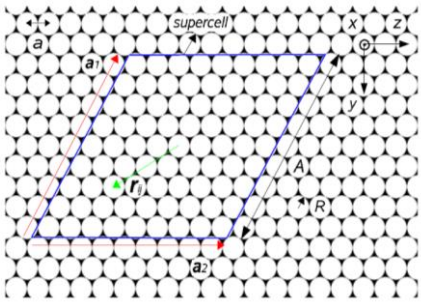


Fig1.3

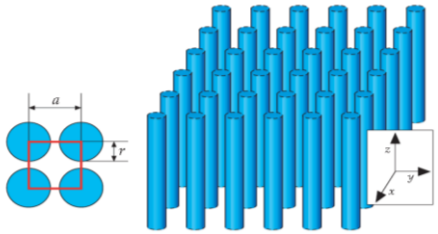
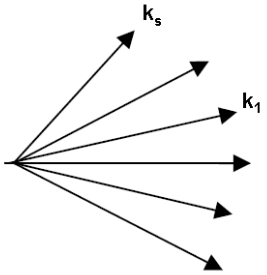


Fig1.4

This graph demonstrate a square array P.C, with a be the distance between the next column and R as the radius. The properties of the 2D photonic crystal are determined by the inter-hole spacing a, the ratio of the radius of the air holes to the spacing r/a and the refractive index of the material slab.

Inverse Fourier transform in time and space



$$\mathrm{H}(\mathbf{r},t) = \iint \mathrm{H}(\mathbf{k}',w)e^{i(\mathbf{k}'\cdot\mathbf{r}-wt)}d\mathbf{k}'dw. \qquad \mathrm{E}(\mathbf{r},t) = \mathrm{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-wt)}$$

Fig1.5 r is the propagation direction.

As we know EM wave H(r, *t*) can be written as an inverse Fourier transform in time and space, which can be thought as a summation of plane waves with infinite numbers of different frequencies(w) and wave vectors(k).It is important that in PC. Cavity are interest to found out the resonant frequency which trap inside.

The Reciprocal Lattice and Brillouin Zone

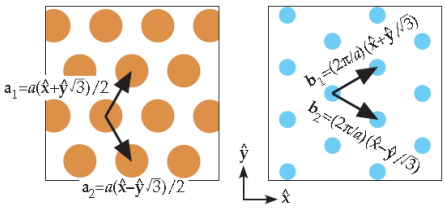


Fig1.6 Fig1.7

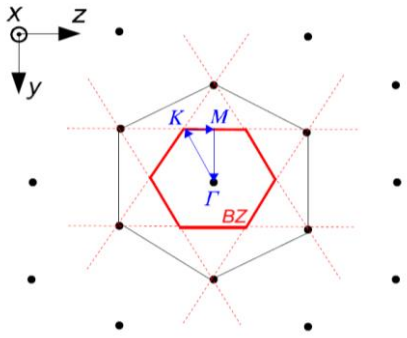
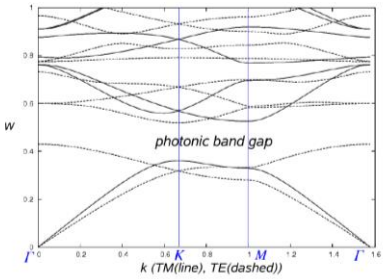
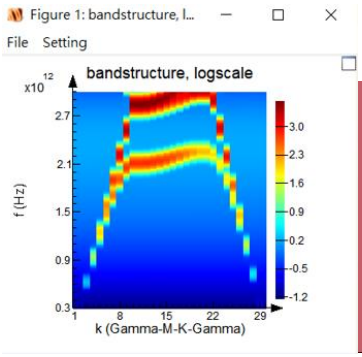


Fig1.8

In here,we want to introduce the reciprocal Lattice and Brillouin zone's to analysis our bandstructure.Leftmost picture show the lattice in time domain, and middle show the k domain which the k vector will be analysis(sweeping) follow the reciprocal space.Since the periodic property we can consider the wave propagate just inside a unit cell which is the Brillouin Zone in some case when we calculate the band-diagram. One of the reason we simulate in the direction of K,M, Γ. A bandgap usually occurs between *every* set of bands, at either the Brillouin zone's edge or its center

Band-diagram and bandgap





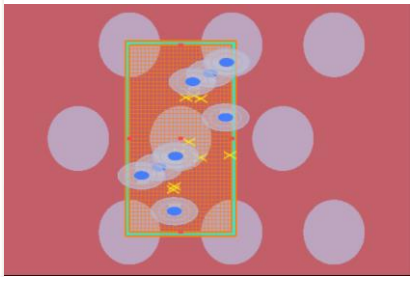
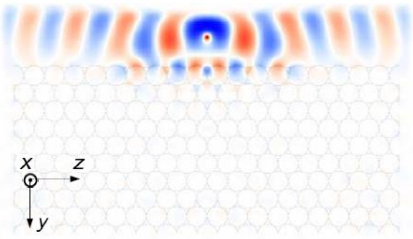
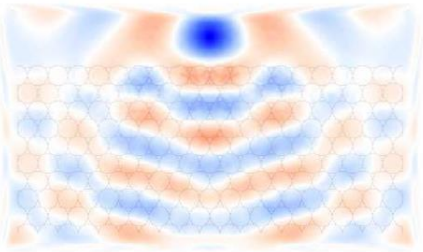


Fig1.9 Fig1.10 Fig1.11

A Band-diagram show frequencies which can exist in the Protonic crystal as when sweeping along the K,M, Γ direction.The middle picture show my simulation of the structure design in right hand side,which show a bandgap exist from ~ 2.4-2.8THz.





(a) (b)

Fig1.12 Fig1.13

Here is the idea to visualize the band-diagram. Let see the leftmost band-diagram. For TE modes for (a) $w = 0.45$, this frequency fall between the bandgap meaning that the photonic crystal do not exixt the EM-wave at normalize frequency 0.45 for any direction in the reciprocal space. Meaning that it can not propagate inside the photonic crystal and $w = 0.20$ frequencies which does show existence of direction and shown in picture (b). 5.

As a conclusion, if we want to design a PC cavity which aims to trap the EM wave, we should creat a bandgap.

Chapter 2.

Methodology & Implementation

Here we will introduce the theory of EM wave in a photonic crystal behavior, and the mathematic simulation methodology FDTD.

2.1 Finite Difference Time Domain Method - FDTD

Finite-difference time-domain (FDTD) is a popular computational electrodynamics modeling method (Yee 1966, Taflove and Brodwin 1975).Let me take a reference of the proof of an research paper to explain. [1] When there is no charge in the 2D PC, but still there will a current density which act as a source, so the Maxwell's equations will be as follows

$$\nabla \cdot \mathrm{D} = 0 \tag{2.5}$$

$$\nabla \cdot \mathrm{B} = 0 \tag{2.6}$$

$$\nabla \times \mathrm{E} = -\frac{1}{c}\frac{\partial \mathrm{B}}{\partial t} \tag{2.7}$$

$$\nabla \times \mathrm{H} = \frac{\mathrm{J}}{c} + \frac{1}{c}\frac{\partial \mathrm{D}}{\partial t} \tag{2.8}$$

When we further derive and extract the component of 2.7 ,2.8

$$\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} = \frac{1}{c}\frac{\partial B_x}{\partial t} \tag{2.9}$$

$$\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} = \frac{1}{c}\frac{\partial B_y}{\partial t} \tag{2.10}$$

$$\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} = \frac{1}{c}\frac{\partial B_z}{\partial t} \tag{2.11}$$

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \frac{J_x}{c} + \frac{1}{c}\frac{\partial D_x}{\partial t} \tag{2.12}$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \frac{J_y}{c} + \frac{1}{c}\frac{\partial D_y}{\partial t} \tag{2.13}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \frac{J_z}{c} + \frac{1}{c}\frac{\partial D_z}{\partial t} \tag{2.14}$$

Assume TE wave which transmit in the xy plane.

$Dx = Dy = Hz = Jx = Jy = 0$ (2.15)

As $D = \varepsilon E$ and $B = \mu H$,we can simplifier the equation as follow

$$-\frac{\partial E_z}{\partial y} = \frac{1}{c}\frac{\partial B_x}{\partial t} \tag{2.16}$$

$$\frac{\partial E_z}{\partial x} = \frac{1}{c}\frac{\partial B_y}{\partial t} \tag{2.17}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \frac{J_z}{c} + \frac{1}{c}\frac{\partial D_z}{\partial t} \tag{2.18}$$

In here we introduce a Yee lattice method in which we can substitute different kind of photonic crystal structure to calculate the E or B field of different time. The main idea is to divide the crystal into mesh, the smaller the mesh, the higher accurate of the calculation.Also for the boundary of different dielectric constant in share the same mesh, it will taking the average and fill in the average dielectric constant in it to simulation.

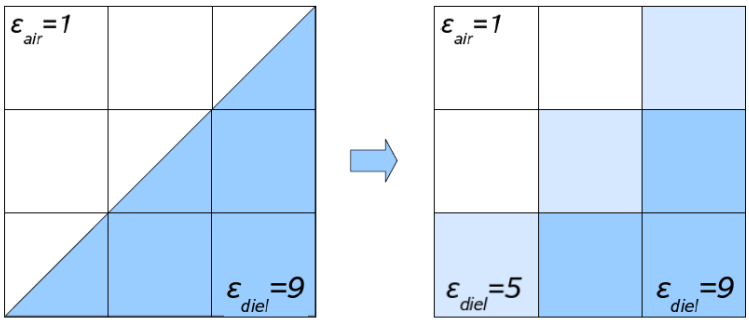


Figure 2.2. A triangular object in a Yee lattice in 2D.

Fig2.1

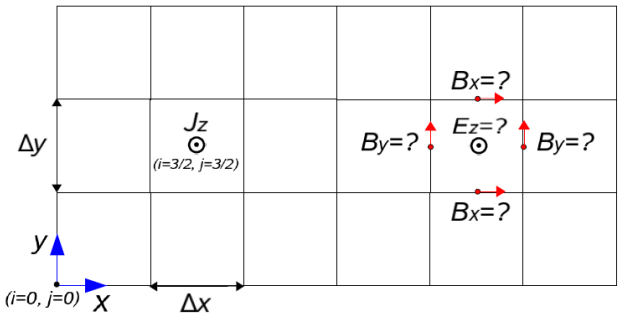


Fig2.2

So the central idea is to divide the simulate region into piece and take calculation, so delta will no longer infinite small

$$-\frac{\Delta E_z}{\Delta y} \approx \frac{\mu}{c}\frac{\Delta H_x}{\Delta t} \tag{2.22}$$

$$\frac{\Delta E_z}{\Delta x} \approx \frac{\mu}{c}\frac{\Delta H_y}{\Delta t} \tag{2.23}$$

$$\frac{\Delta H_y}{\Delta x} - \frac{\Delta H_x}{\Delta y} \approx \frac{J_z}{c} + \frac{\epsilon}{c}\frac{\Delta E_z}{\Delta t} \tag{2.24}$$

At last, we rearrange the equation, we will get a recursive equation, meaning that once the source J is excite, E field will be derive and D field can also be derive recursively in different i,j position and different time.

$$E_z^{k+\frac{1}{2}}(i+\frac{1}{2},j+\frac{1}{2}) \approx E_z^{k-\frac{1}{2}}(i+\frac{1}{2},j+\frac{1}{2}) - \frac{\Delta t J_z^k(i+\frac{1}{2},j+\frac{1}{2})}{\epsilon(i+\frac{1}{2},j+\frac{1}{2})} + \frac{c\Delta t}{\Delta x}\frac{H_y^k(i+1,j+\frac{1}{2}) - H_y^k(i,j+\frac{1}{2})}{\epsilon(i+\frac{1}{2},j+\frac{1}{2})} - \frac{c\Delta t}{\Delta y}\frac{H_x^k(i+\frac{1}{2},j+1) - H_x^k(i+\frac{1}{2},j)}{\epsilon(i+\frac{1}{2},j+\frac{1}{2})} \quad (2.28)$$

$$H_x^{k+1}(i+\frac{1}{2},j) \approx H_x^k(i+\frac{1}{2},j) - \frac{c\Delta t}{\Delta y}\frac{E_z^{k+\frac{1}{2}}(i+\frac{1}{2},j+\frac{1}{2}) - E_z^{k+\frac{1}{2}}(i+\frac{1}{2},j-\frac{1}{2})}{\mu(i+\frac{1}{2},j)} \quad (2.29)$$

$$H_y^{k+1}(i,j+\frac{1}{2}) \approx H_y^k(i,j+\frac{1}{2}) + \frac{c\Delta t}{\Delta x}\frac{E_z^{k+\frac{1}{2}}(i+\frac{1}{2},j+\frac{1}{2}) - E_z^{k+\frac{1}{2}}(i-\frac{1}{2},j+\frac{1}{2})}{\mu(i,j+\frac{1}{2})} \quad (2.30)$$

Chapter 3.

Photonic crystal cavity structural Design (One dimensional)

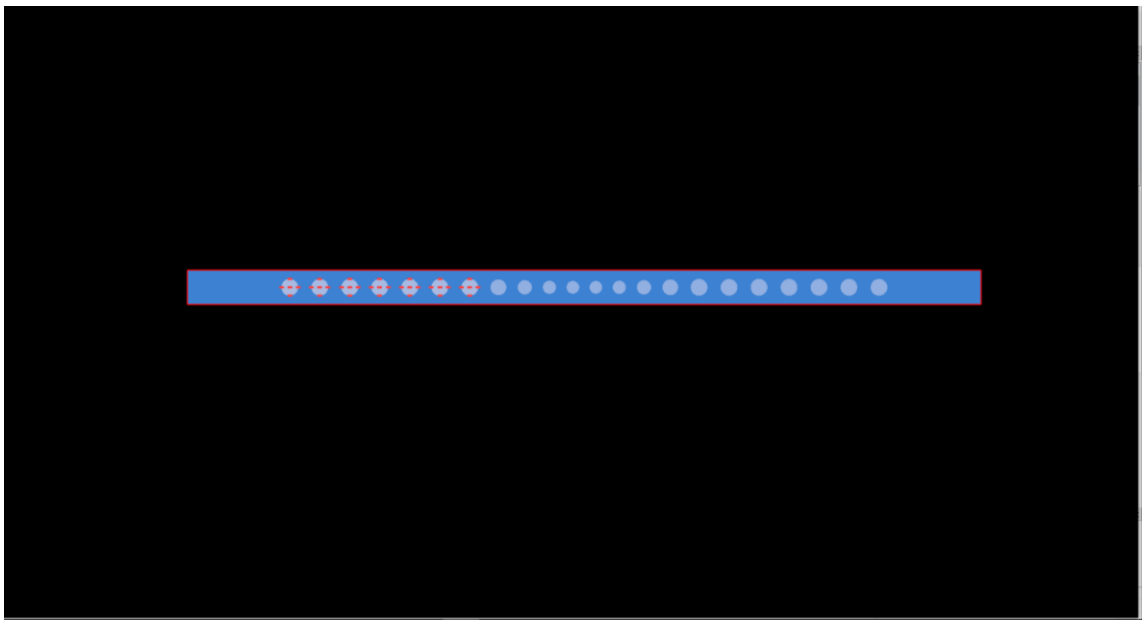


Fig3.1

There are 7 periodic air column in side the substite which radius (Rmax) are constants and increment of displacement are constants.

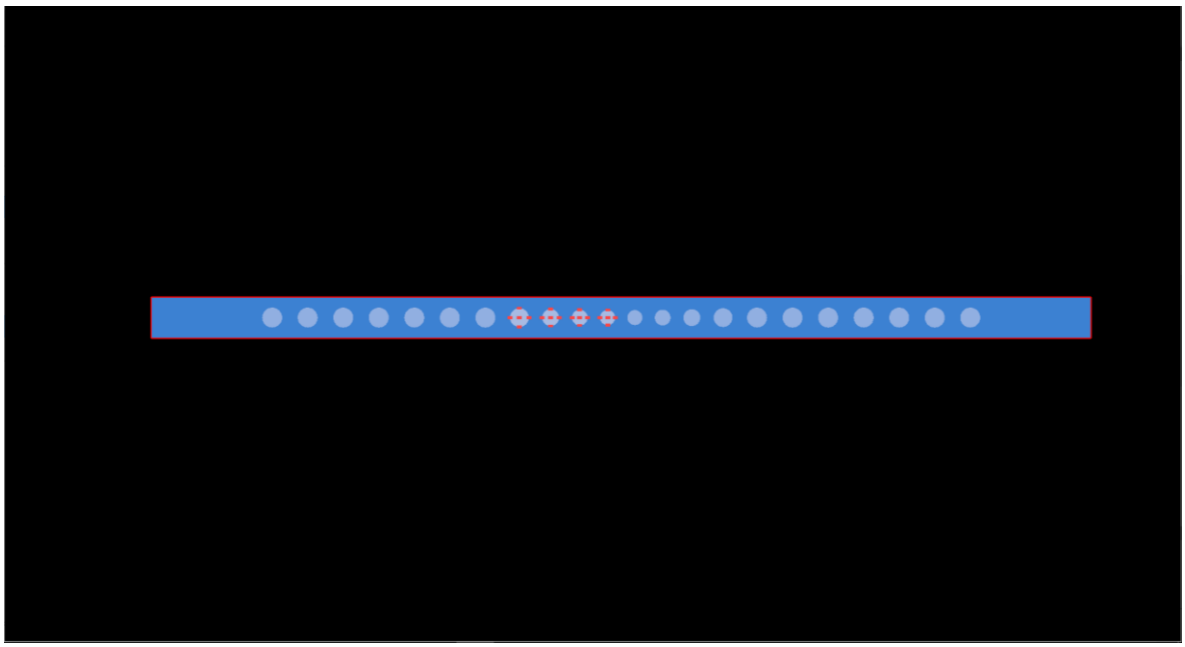


Fig3.2

There are 4 air column in the taper section which the radius follow the parabolic function aN^2+b = rN. The radius (Rmin ,N = 1) is smallest near the center, and larger when away the center(N = 2,3,4). Also the center point of cavity act as the mirror image of the structure(Symmetric). Once Rmin is known, Rmax and by setting the total number of air column in taper section. The parabolic function is found given by a,b you solve.

Rmin = a(1)^2+b

Rmax =a(4)^2+b

rN = 0.112815885N^2 +8.82224946\

In my structure,I have fix the taper section as 4 and Rmax ,by sweeping different size of Rmin, the radius of section will follow the function for optimizations .Also the position (X) of tapers section follow the function $X = \frac{N^4}{800000} + \frac{1649N^3}{1200000} + \frac{47N^2}{800000} + \frac{391723N}{1200000} + \frac{16287}{100000}$

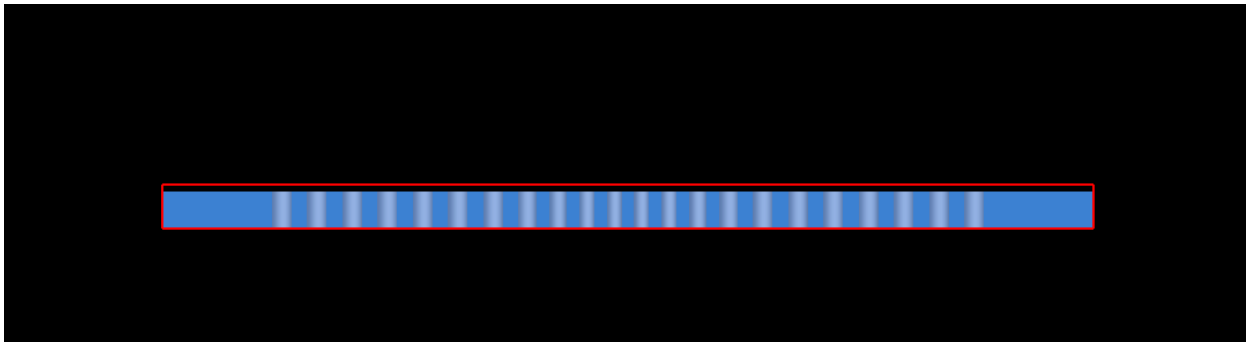


Fig3.3

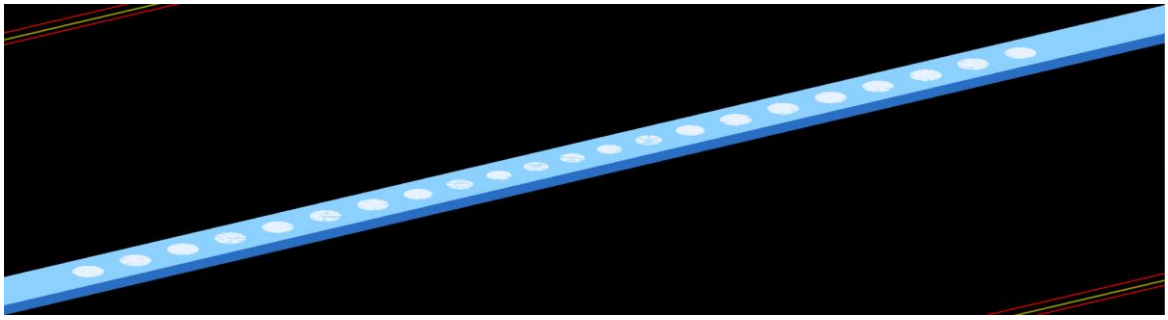


Fig3.4

Chapter 4

Simulation result

Parameter of Waveguide

Width(μm)	Length(μm)	Thickness(μm)	Refractive index
48.3499	1100	21.274	3.4175

Parameter of Air Hole

Index	Position(μm)	Radius(μm)
N = 1 (Rmin)	15.9555	8.79969
N = 2	48.4708	9.1607
N = 3	82.5981	9.88273
N = 4	119.142	10.9658
N = 5 (Rmax)	158.91	11.6427
N = 6	200.491	11.6427
N = 7	242.072	11.6427
N = 8	283.653	11.6427
N = 9	325.234	11.6427
N = 10	366.815	11.6427

N = 11	408.396	11.6427

Simulation Results

Resonant Frequency(THz)	2.10681 λ=142.395375μm
Q factor (Lossless case)	2.71009x10^6
Q factor (Loss case,FDTD material database)	3275.79
Q factor (Loss case, high-resistivity float-zone single-crystal silicon)	64485.1

Q factor (Loss case,FDTD material database)

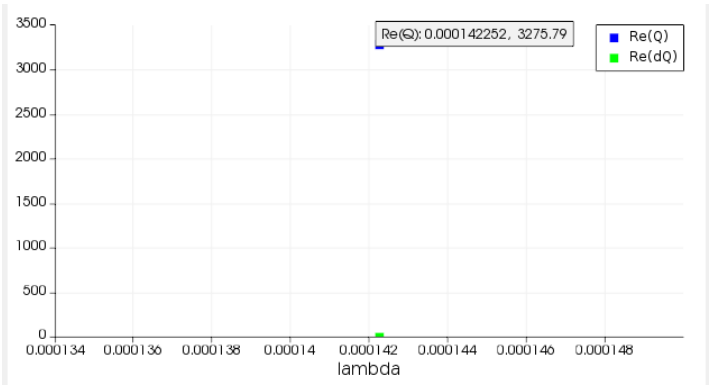


Fig3.5

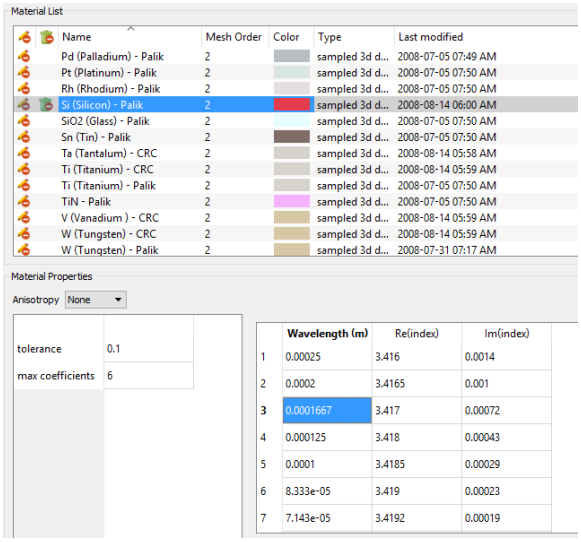


Fig3.6

As the resonant wavelength is 0.000142395375m.Is fall between 0.0001667m to 0.000125m. By using linear approximation due to dx is small,the refractive index is sex to 3.4175. As we see that the imaginary part of RI is 0.00072 and 0.00043. Due to the reason that the material database silicon in the Lumerical FDTD have a high loss on the range of terahertz frequency. Later on I have search about some high resistive silicon database market.

Q factor (Lossless case)

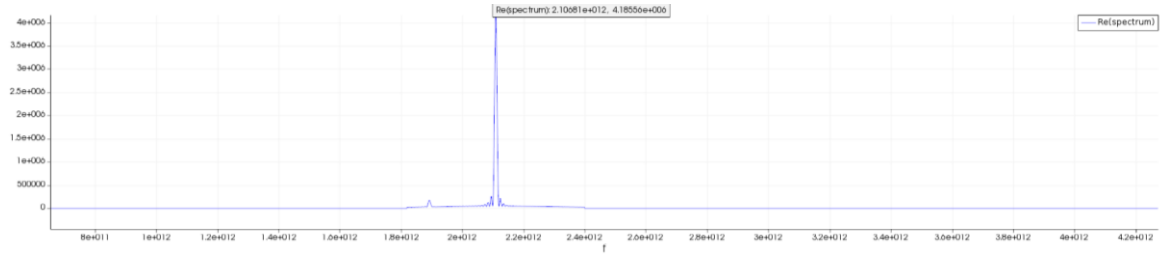


Fig3.7

18.

Sweep 1 for Radius(min) Mesh accuracy = 2

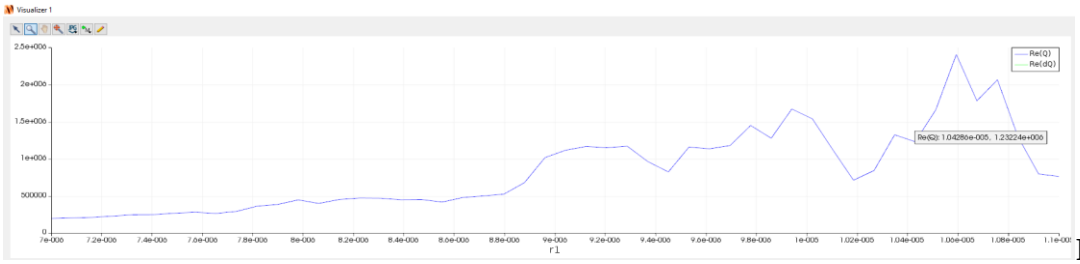


Fig3.8

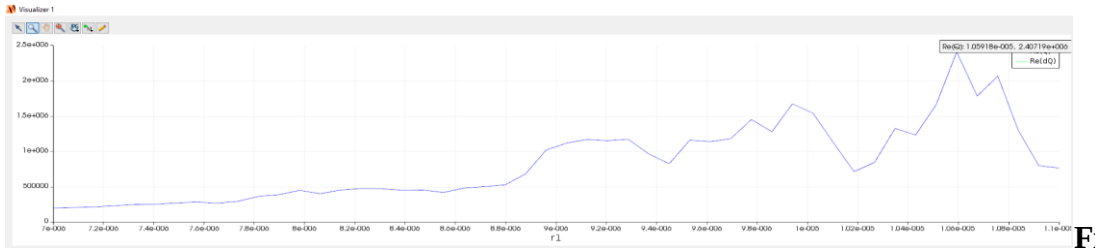


Fig3.9

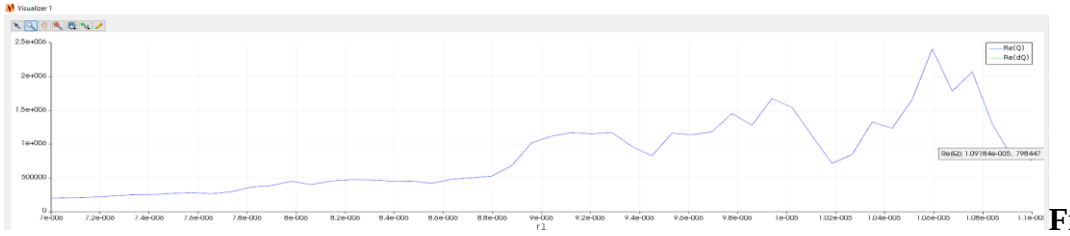


Fig3.10

Sweeping Rmin from 7um to 11um for 50 slices, as we see there are a peak betweenthese two frequency but there are a local minimum.So there indicate it may be further optimize.

Sweep 2 for Radius(min) (10.4286um – 10.9184um) ,20 slice,Mesh accuracy = 2

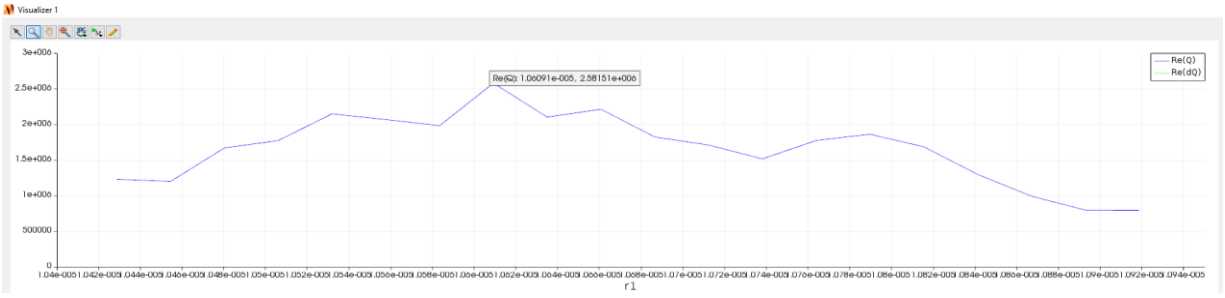


Fig3.11

Rmin = 10.6091um , Q = 2.58151 x 10^6

Mesh accuracy = 3

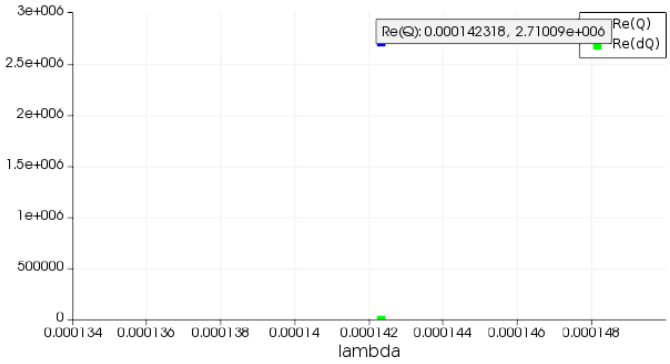


Fig3.12

Rmin = 10.6091um ,Q = 2.71009 x 10^6

E-Field

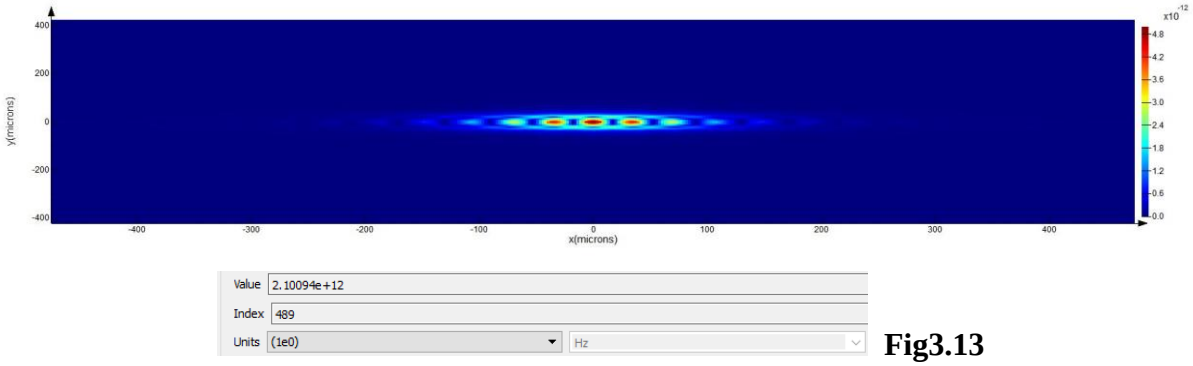


Fig3.13

Q factor (Loss case, high-resistivity float-zone (FZ) single-crystal silicon)[2]

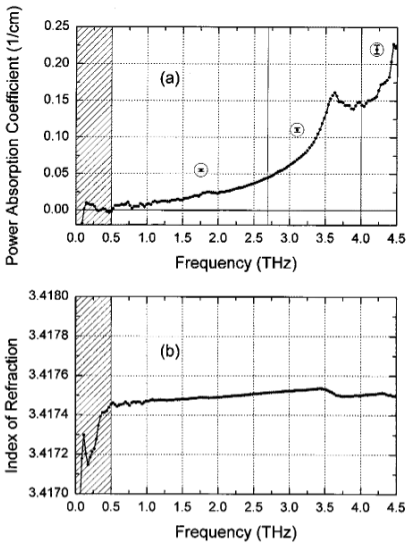


Fig3.14

when Frequency is 2THz Absorption coefficient ~0.025cm^-1 , n~3.4175,by equation$\alpha = \frac{4(\mathrm{pi})(\kappa)}{\lambda}$ related to absorption(α) and extinction coefficient κ,where is the imaginary part of refractive index, $\underline{n} = n + i\kappa$,$\alpha = \frac{4(\mathrm{pi})(\kappa)}{\lambda} \rightarrow \frac{0.025(\mathrm{m}\text{^}-1)}{10\text{^}-2} = \frac{4(\mathrm{pi})(\kappa)}{142.395375\mathrm{x}10^{-6}\ \mathrm{m}}$->

κ = 0.000028328

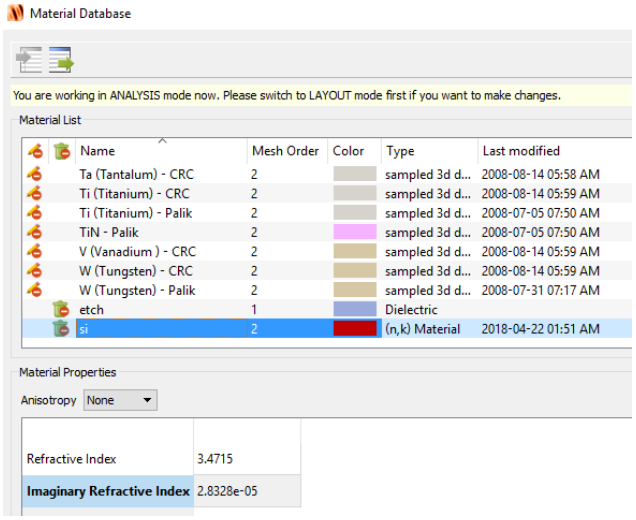


Fig3.15

By adding a new material in database

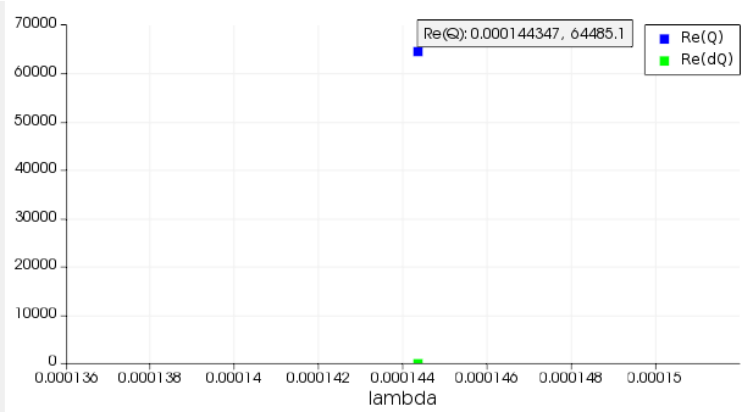


Fig3.16

Chapter 5　Simulation Setup

5.1 Photonic crystal structural setting

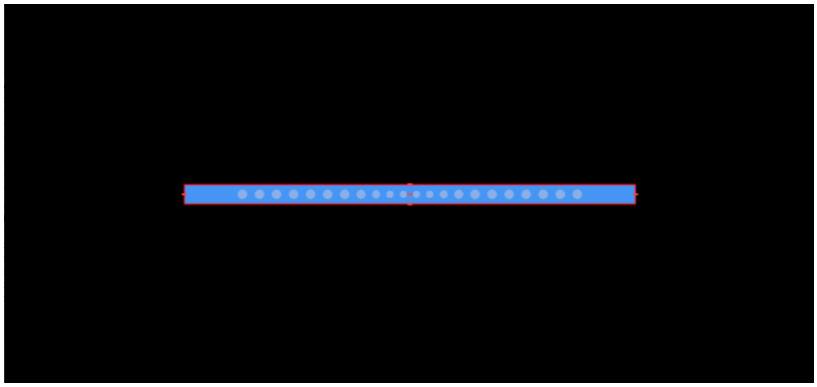


Fig5.1

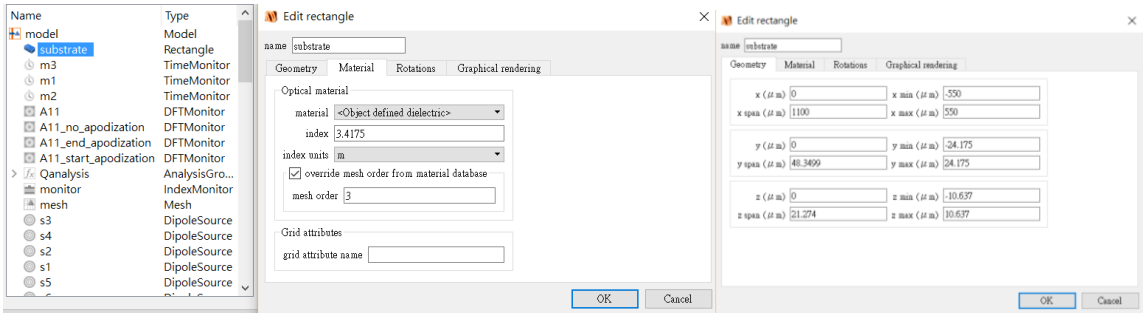


Fig5.2

Creat the substrate and set the length, height, thickness,refractive index

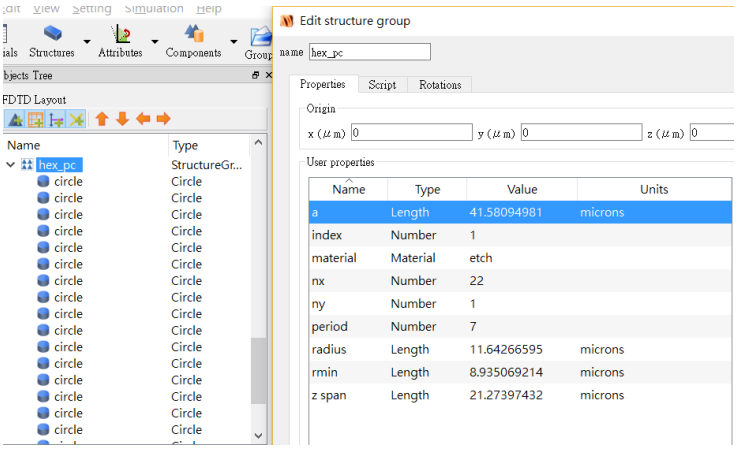


Fig5.3

Set the structure group of air column follow the parobolic function by modify the Script and adding parameter like rmin. Set etch as override the substrate.Nx is the total number of air column,radius is set to ~11.6426 which is Rmax,new parameter rmin is added for sweeping of parabolic function,period is set as 7 represent the periodic part. Also we can utilize some scaling properties on Maxwell equation which help us to transform the high profermance on one scale to another but it doesn't promise but as a reference for further tuning.

$\lambda 1/\lambda 2 \sim a1/a2$, $r1/a1 \sim r2/a2$, $h1/a1 \sim h2/a2$

Sweeping

rmin =2um

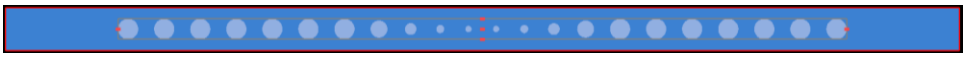


Fig5.4

rmin =4um

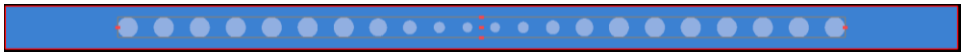


Fig5.5

rmin =8um

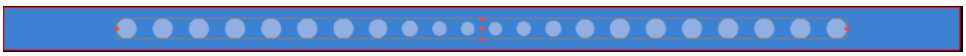


Fig5.6

rmin =10um



Fig5.7

5.2 FDTD Setting

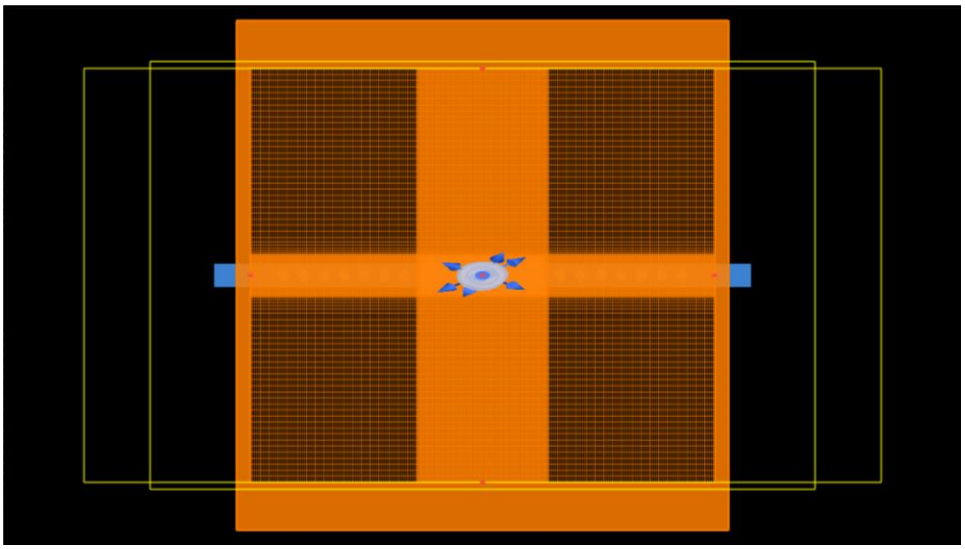


Fig5.8

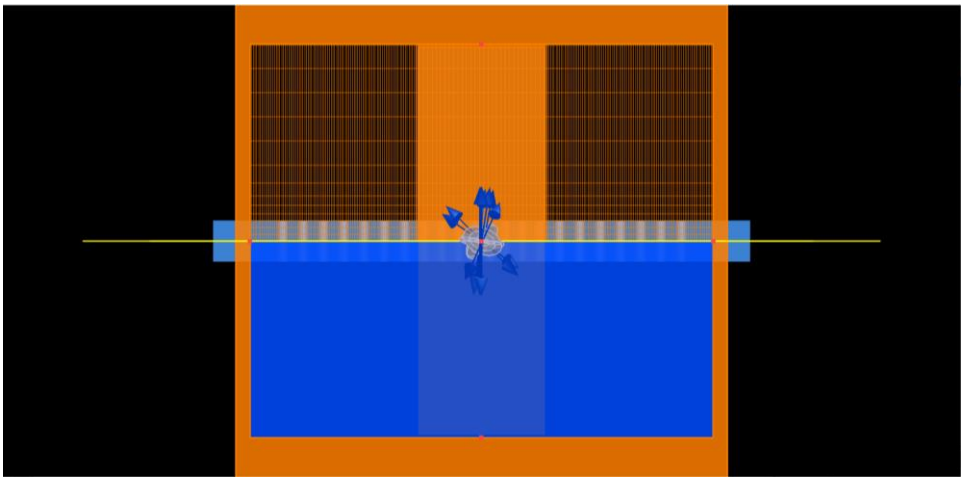


Fig5.9

5.3 Perfectly matched layer (PML) absorbing boundaries

It is important to set the absorbing boundary of the FDTD simulation, when the dipole generate the signal in the cavity, all the EM-wave leak to outside should spread to the infinity side without reflection and effect the simulation of the structure.

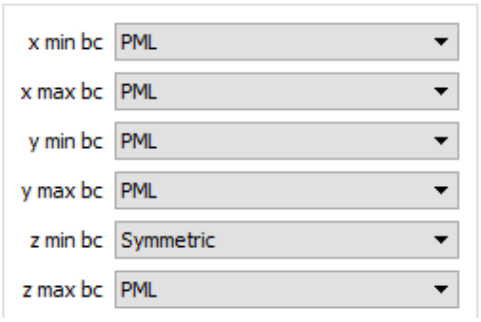


Fig5.10

The FDTD simulation region should be larger than ~2 wavelength both in width and height, otherwise it will effect the simulate of the EM wave in which we only desire to absorb the wave lackage but not the cavity.It is also keep remind the (PML) absorbing boundaries should set with in the substrate (x direction) avoiding the leakage of EM wave reflect back to the cavity.

Dipole source

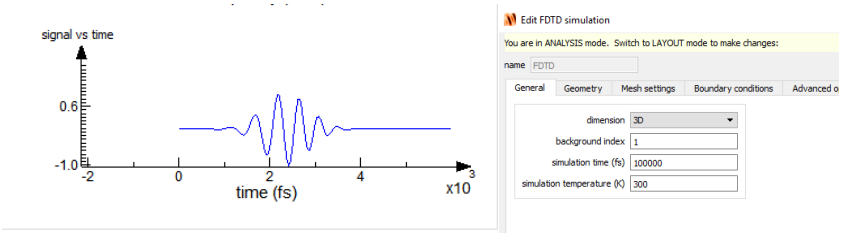


Fig5.11

3D simulation must be perform on the model as the accuracy for real case.As we set the dipole group later, the signal generate is end at around 5000fs,in here we set the simulation time as 100000 which is 20 time of the source duration,which should be large enough to visualize whether the certain frequency of EM- wave are trap in a position but not lost its energy quickly, if we set it too short, we can not distinguish the structure design is a cavity

which the EM wave stay for a long time, but if we set it too long, the simulate time will be much longer, so we need to have a balance on it.

5.4 Dipole Cloud Source and Time monitor

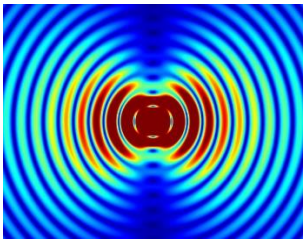


Fig5.12

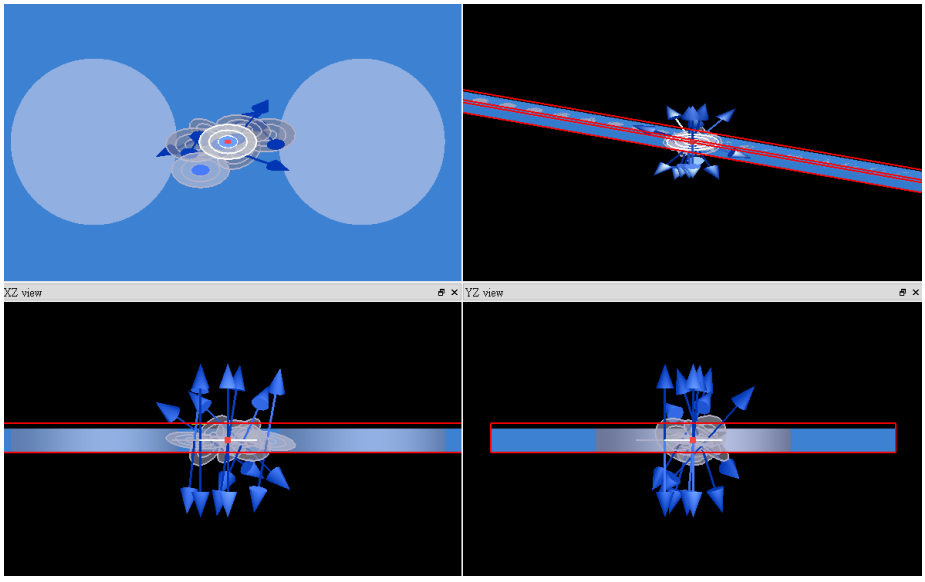


Fig5.13

Dipole cloud are set as the signal generation in all direction by setting theta, thi in degree and placing at different position inside the cavity to excite all modes and avoiding blind spot.

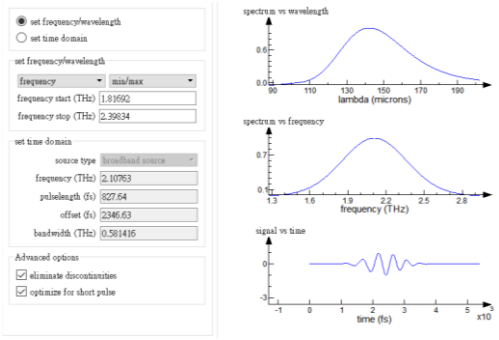


Fig5.14

Setting the global source and the start , stop frequency is being modulate, being narrower after further optimizations to filter only the peak resonant frequency range to excite desire mode. Time monitors cloud are also create in the similar manner by setting randomly distribute inside the crystal cavity simulate region to collect fields over time. After the dipole sources excite and it will continuous to simulate for the time we set before, destructive interference occurs when time gone and most of the fields will decay and remain the mode which stay strong and trap inside the cavity, and they will be detected by the time monitor.

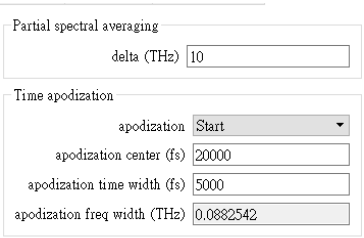


Fig5.15

Apodization which apply applies a window function to the simulation fields E(t) before the monitor performs its Fourier transform of E(t) to obtain E(w),in here we use start Apodization which can be used to ignore all transients which occur near the start of the simulation.

In order filtering away short lived transients that occur when a system is excited with a dipole source,when studying high Q systems that decay very slowly.

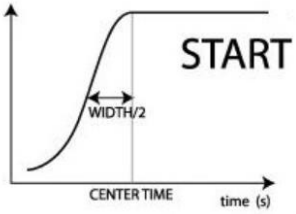


Fig5.16

Second by Fourier transform the time signal receive for each monitors to get the frequency domain of those exist mode, then by summation all of the set of Fourier transform spectrum, we will able to find out all the resonant frequencies, even through one of the monitor located at a node of mode which can detect the mode that actually exist in the structure will shown in other monitor. Then the program will repeat the produce by sweep through all different k vector to get all set of data. By setting the number of band you are desire, the program will plot the largest frequency peaks for each k vector .[4]

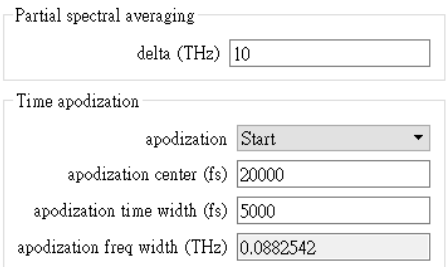


Fig5.17

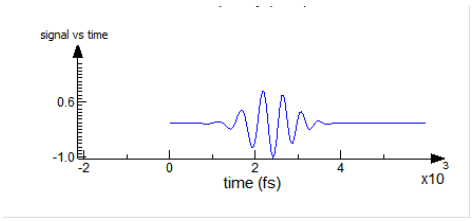


Fig5.17

Signal source is 5000fs and the apodization center are set to 20000fs further after it.

And also We can visualize the E-Field in the certain frequency.

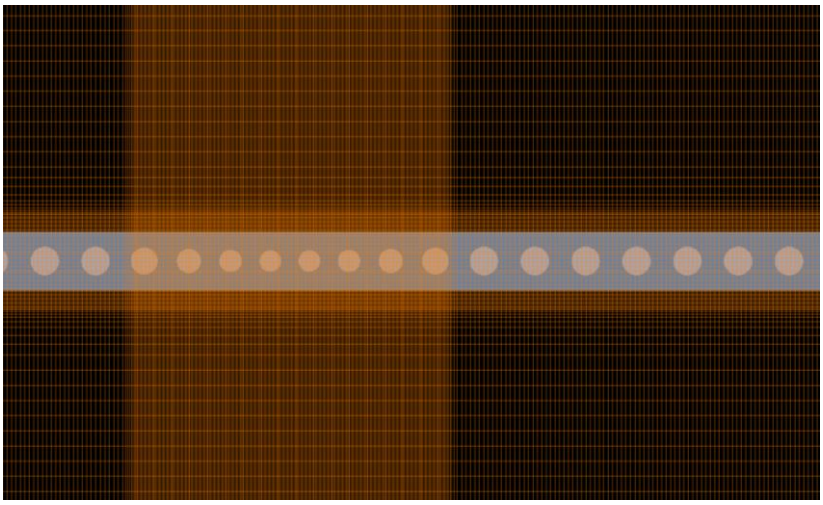


Fig5.17

As we have set auto non uniform of mesh type in FDTD, as we see more non-uniform mesh is focus on the structure and also adding a mesh on more focus on the taper region for accuracy and balance of simulation time as we perform sweeping, the time consume is important, as empirical experience around (λ/15) of mesh size is effectively accurate, we further slicing the mesh , the simulation result will be similar, but the time consume will increase.

Chapter 6. Other structure trial

Structure 1

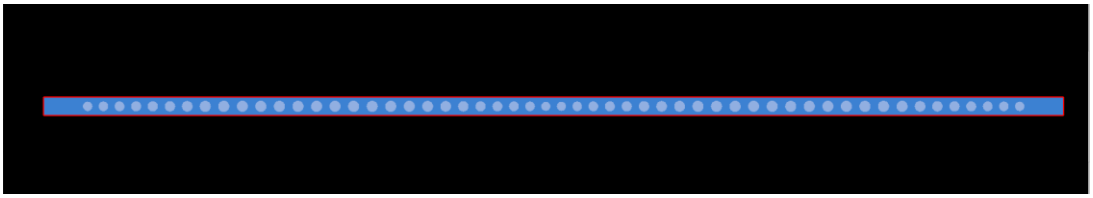
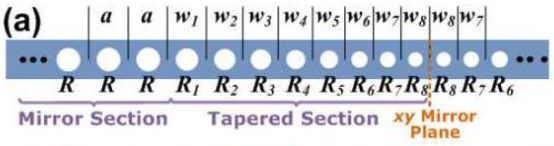


Fig6.1

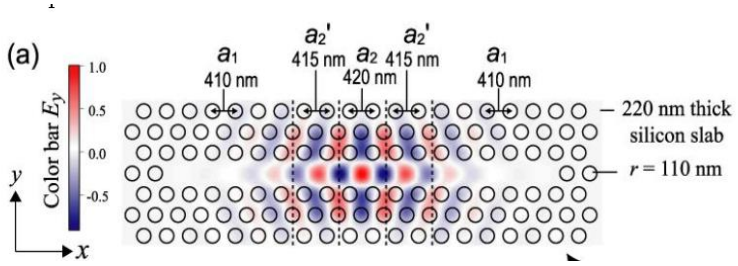


[3]

Fig6.2

By first following some of the thesis and rescale to Terahertz range at the early stage of semester 2. Symmetric at the center of cavity, also both end taper are also symmetric follow the ratio Rk/wk fixed where Rk is the radius of the taper and wk is the displacement,with Rmin and Rmax of two degrees of freedom.

Structure 2



[4]

Fig6.3

Displacement of inner region

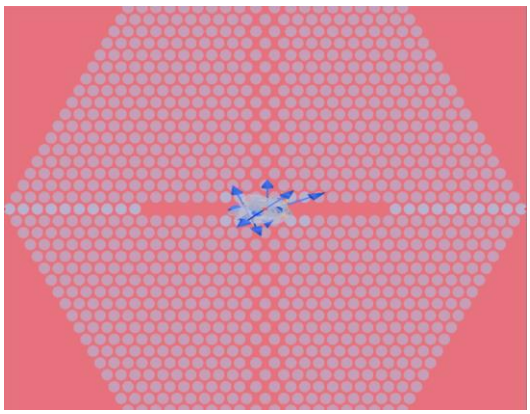


Fig6.4

Displacement of middle region

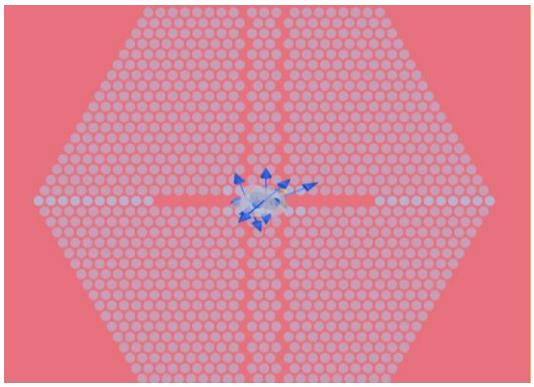


Fig6.5

Displacement of Inner + middle region

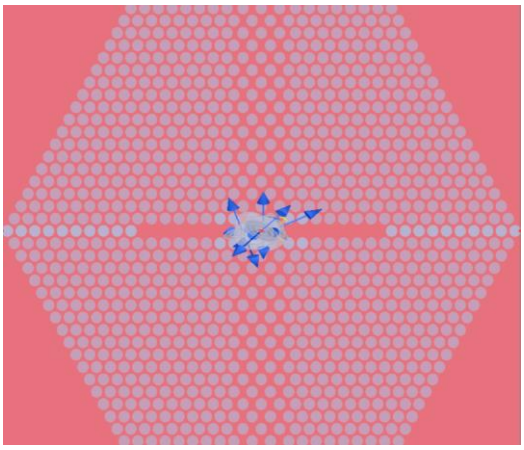


Fig6.6

E-field

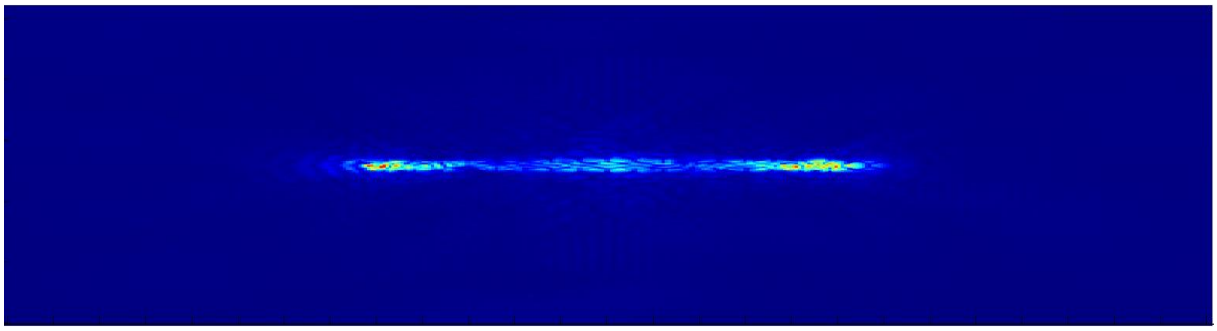


Fig6.7

As I have notice the problem of material loss is too high in the later stage, the following best simulation result using FDTD lumerical's material database,Q ~3000

Chapter 6

Conclusions

As we see the E-field of one dimensional structure shows EM-wave is welly confine inside the cavity, the advantage of 1D PC cavity is compact size and the highly confinement and low mode volumes .The structure can be use as gas sensing, the change of concentration of gas indicate the change of background refractive index,the cavity will result in a different resonant frequency, When RI increase, the resonant wavelength shifts to longer wavelength(red-shift),sensitivity is govern by sensitivity $S = \Delta\lambda / \Delta n$,

References

[1] Adem Enes Erol, “Frequency Splitting With Two Dimensional Triangular Photonic Crystal”, Master of Science Degree of in Physic, Graduate School of Engineering and Sciences of˙Izmir Institute of Technology,2009

[2] Jianming Dai, Jiangquan Zhang, Weili Zhang, and D. Grischkowsky
Terahertz time-domain spectroscopy characterization of the far-infrared absorption and index of refraction of high-resistivity, float-zone silicon ,2004
School of Electrical and Computer Engineering, Oklahoma State University, Stillwater, Oklahoma

[3] Yinan Zhang,∗ Murray W. McCutcheon, Ian B. Burgess, and Marko Loncar, Ultra-high-Q TE/TM dual-polarized photonic crystal nanocavities, School of Engineering and Applied Sciences, Harvard University, Cambridge, MA 02138, USA,May 2009

[4] Yuki Taguchi,1 Yasushi Takahashi,2,3,* Yoshiya Sato,1 Takashi Asano,1 and Susumu Noda1, Statistical studies of photonic heterostructure nanocavities with an average Q factor of three million,
1 Department of Electronic Science and Engineering and Photonics and Electronics Science and Engineering Center, Kyoto University, Kyoto 615-8510, Japan

2 Nanoscience and Nanotechnology Research Center, Research Organization for the 21st Century, Osaka Prefecture University, Sakai, Osaka 599-8570, Japan
3 PRESTO, Japan Science and Technology Agency (JST), 4-1-8 Honcho Kawaguchi, Saitama 332-0012, Japan,2011

Ahmad Rifqi Md Zain, One-dimensional Photonic,Crystal / Photonic Wire Cavities Based on Silicon-on-Insulator (SOI), Department of Electrical and Electronics Engineering University of Glasgow, April 2009

Seyed Hamed Mirsadeghi,Optical Trapping Using a Photonic Crystal Cavity: Design and Sample Fabrication, Faculty of Graduate Studies (Physics),The University of British Columbia, April, 2010

Lumerical,Simulation setup
Internet:https://kb.lumerical.com/en/diffractive_optics_pc_simulation_methodology.html
https://kb.lumerical.com/

Photonic crystals _ molding the flow of light_J. D. Joannopoulos, S. G. Johnson, J.N. Winn, R.D. Meade -Princeton univ press, 2008

Integrated photonics Fundamentals Gin´es Lifante Universidad Aut´onoma de Madrid, Spain, 2003

Kazuisao Tsuruda, Masayuki Fujita, Asako Suminokura, Masahiro Yata,
Toshikazu Mukai, and Tadao Nagatsuma,Terahertz-wave Integrated Circuits Based on Photonic Crystals,Graduate School of Engineering Science, Osaka University,Japan 2015

N. Born, M. Scheller,M. Koch,and J. V. Moloney, Cavity enhanced terahertz modulation, College of Optical Sciences, University of Arizona,USA
Faculty of Physics and Material Sciences Center, Philipps-Universitat Marburg, Germany,2014

Cristo M. Yee and Mark S. Sherwin, High- terahertz microcavities in silicon photonic crystal slabs, Department of Physics, Institute for Quantum and Complex Dynamics, University of California, USA,2009

Ryoma Kakimi, Masayuki Fujita*, Masaya Nagai, Masaaki Ashida and Tadao Nagatsuma, Capture of a terahertz wave in a photonic-crystal slab, Graduate School of Engineering Science, Osaka University , Japan,2014
Kazuisao Tsuruda, Masayuki Fujita,* and Tadao Nagatsuma, Extremely low-loss terahertz waveguide based on silicon photonic-crystal slab, Graduate School of Engineering, Osaka University,japan,2015

L. Bingham and D. Grischkowsky, Terahertz two-dimensional high-Q photonic crystal waveguide cavities, School of Electrical and Computer Engineering,USA,2008

Masayuki Fujita*a and Tadao Nagatsuma, Photonic crystal technology for terahertz system integration, aGraduate School of Engineering Science, Osaka University,Osaka,Japan,2017